

Data Warehousing and Business Intelligence

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Step 1: Data source for the assignment 2 Data Warehouse Design

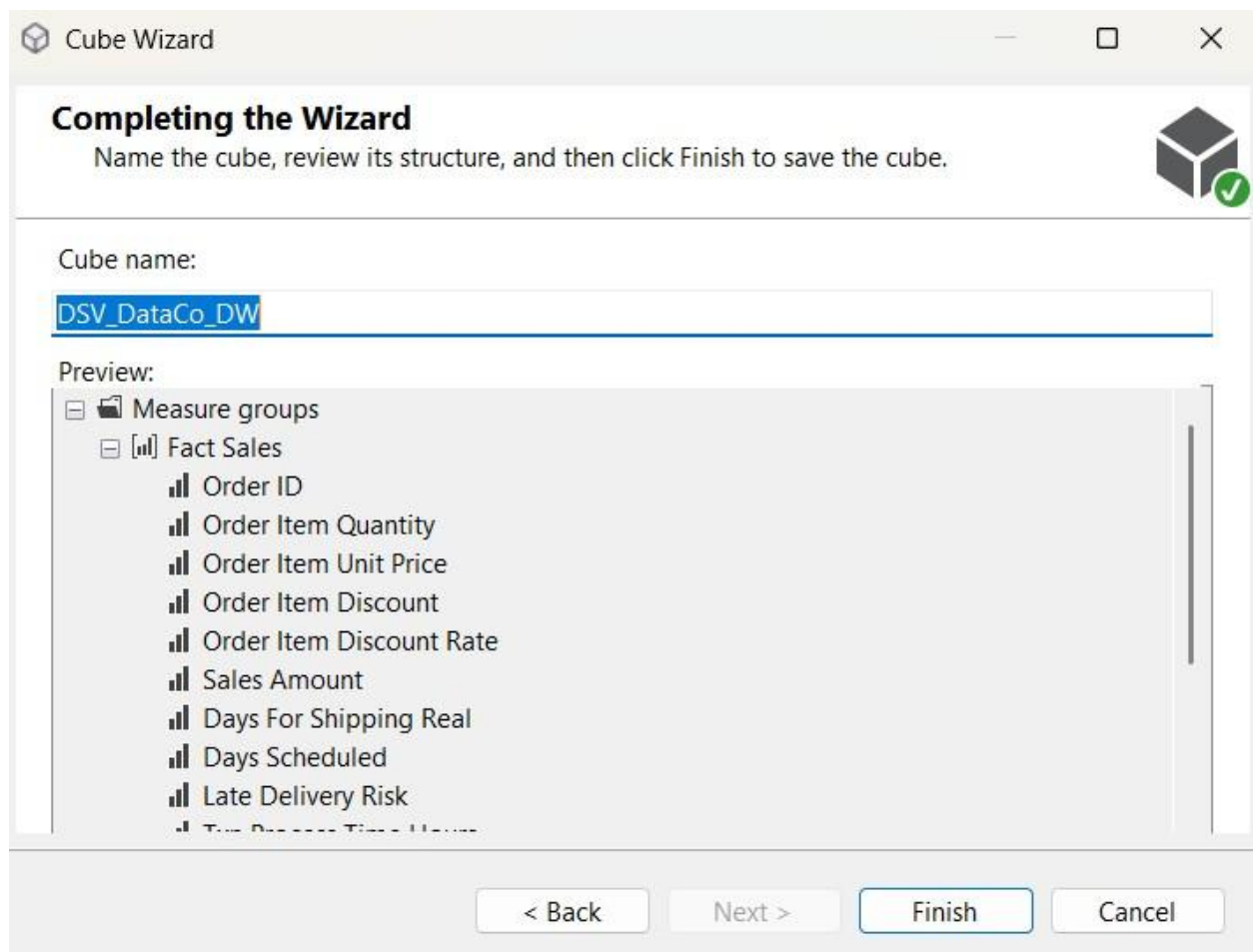
The data source used for Assignment 2 is the DataCo Smart Supply Chain data warehouse (DataCo_DataWarehouse) that was fully designed and populated in Assignment 1. This SQL Server data warehouse implements a star schema dimensional model and serves as the analytical foundation for the SSAS cube and Power BI reports developed in this assignment.

Table	Type	Description	Key Columns
Fact_Sales	Fact Table	Captures order item level sales transactions.	OrderFactKey, DateKey, ProductKey, CustomerKey
Dim_Product	Dimension	Product master data with SCD Type 2 tracking. Contains 118 products across 11 departments.	ProductKey (PK), ProductID, ProductName, CategoryName, DepartmentName, StartDate, EndDate, IsCurrent
Dim_Customer	Dimension	Customer demographics with SCD Type 2.	CustomerKey (PK), CustomerID, CustomerFnam, CustomerLnam, CustomerSegment, CustomerCity, CustomerState, CustomerCountry, StartDate, EndDate, IsCurrent
Dim_Date	Dimension	Conformed date dimension supporting time-based analysis	DateKey (PK), FullDate, Year, Quarter, Month, MonthName, Day, WeekdayName, IsWeekend

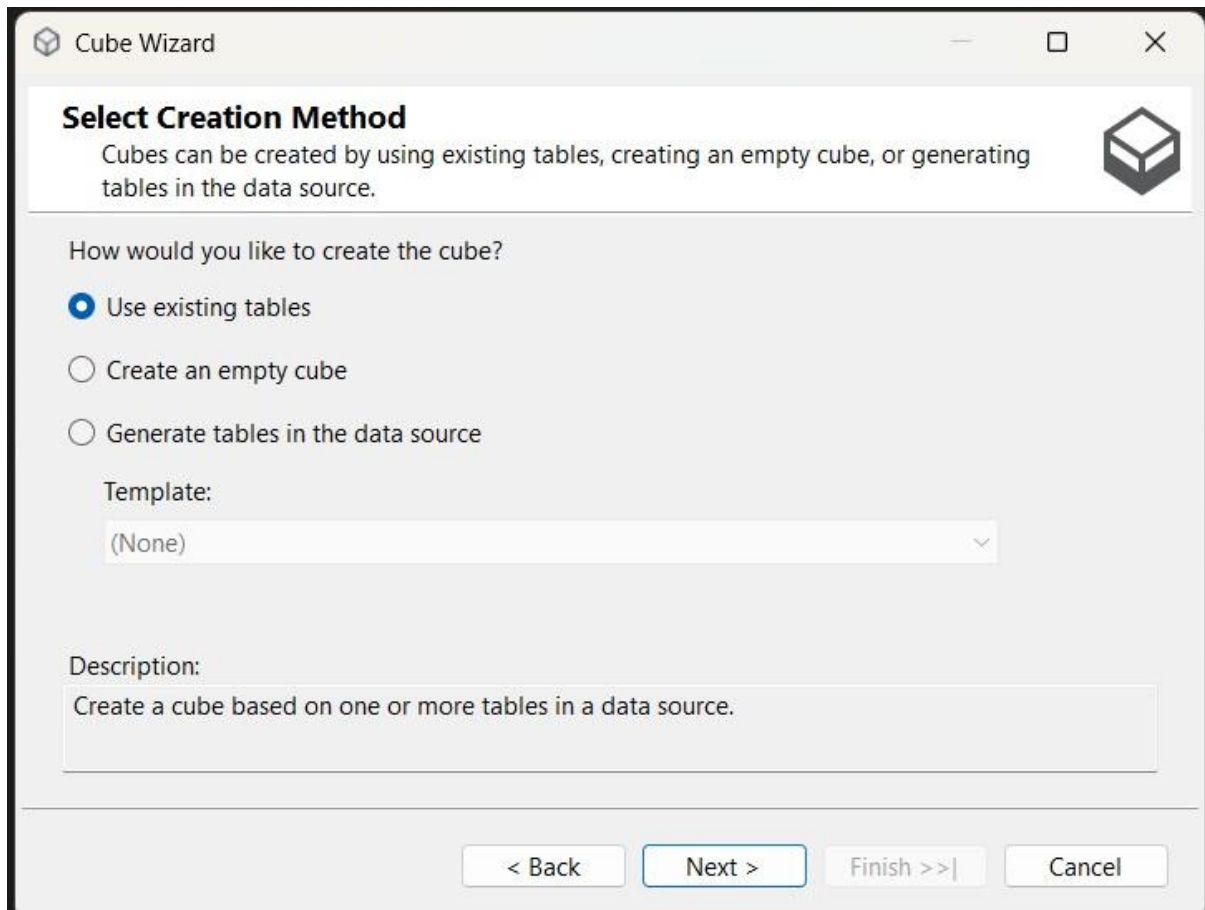
Step 2: SSAS Cube implementation

An SSAS (SQL Server Analysis Services) multidimensional cube named Cube_DataCo was created using Visual Studio 2022 with the Analysis Services project template. The cube was built on top of the DataCo_DataWarehouse and includes all three dimensions with three custom hierarchies.

1. Created SSAS Project - Launched SQL Server Data Tools (SSDT) and created a new "Analysis Services Multidimensional Project".
2. Established Data Source - Configured a connection to the existing data warehouse database using Windows authentication.
3. Built Data Source View - Imported all fact and dimension tables from the DW and manually established relationships between them.

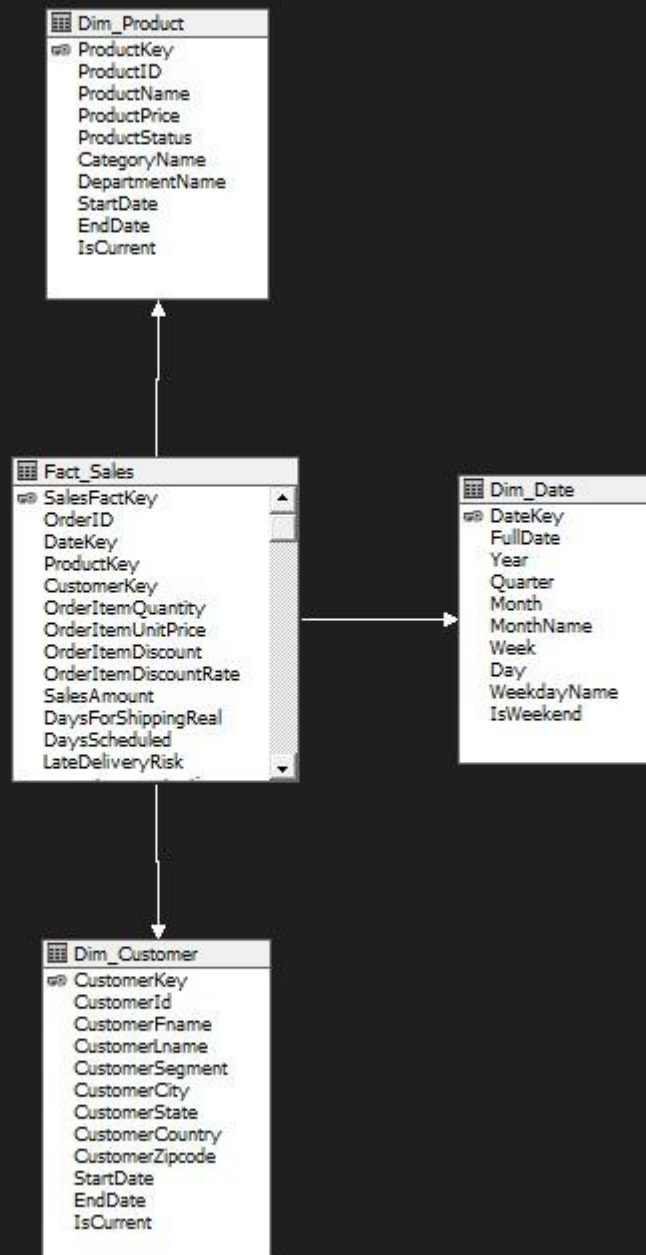


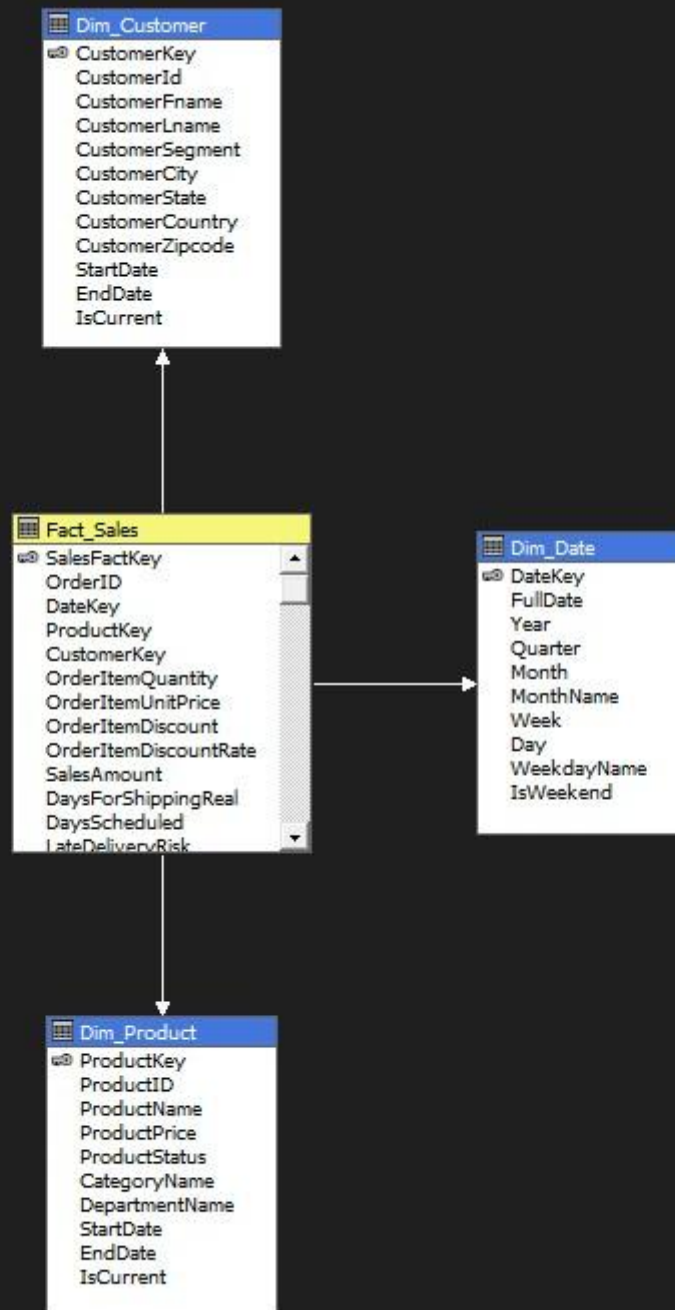
4. Initiated Cube Creation - Used the Cube Wizard with "Use existing tables" option and selected the fact table as the measure group.



The screenshot shows a window titled "Cube Wizard" with a close button (X) in the top right corner. The main heading is "Select Creation Method". Below the heading, a descriptive text states: "Cubes can be created by using existing tables, creating an empty cube, or generating tables in the data source." To the right of this text is a small 3D cube icon. The question "How would you like to create the cube?" is followed by three radio button options: "Use existing tables" (which is selected), "Create an empty cube", and "Generate tables in the data source". Below these options is a "Template:" label and a dropdown menu currently showing "(None)". Further down is a "Description:" label and a text input field containing the text "Create a cube based on one or more tables in a data source." At the bottom of the window, there are four buttons: "< Back", "Next >" (highlighted with a blue border), "Finish >>|", and "Cancel".

5. Incorporated Dimensions - Added all dimension tables to create the star schema structure.
6. Enhanced Dimensions - Edited each dimension to include all attributes and set appropriate key attributes.



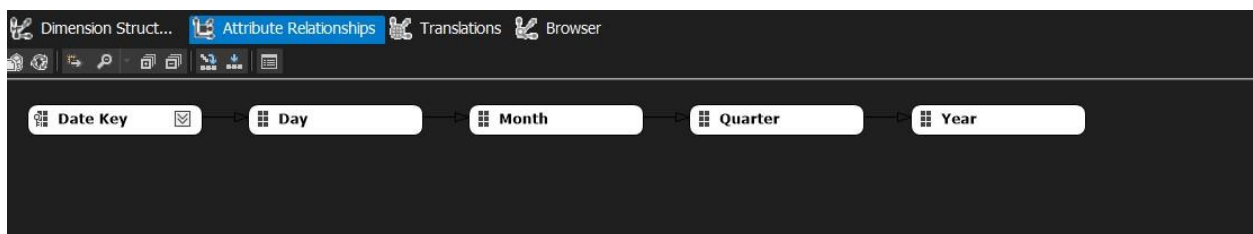
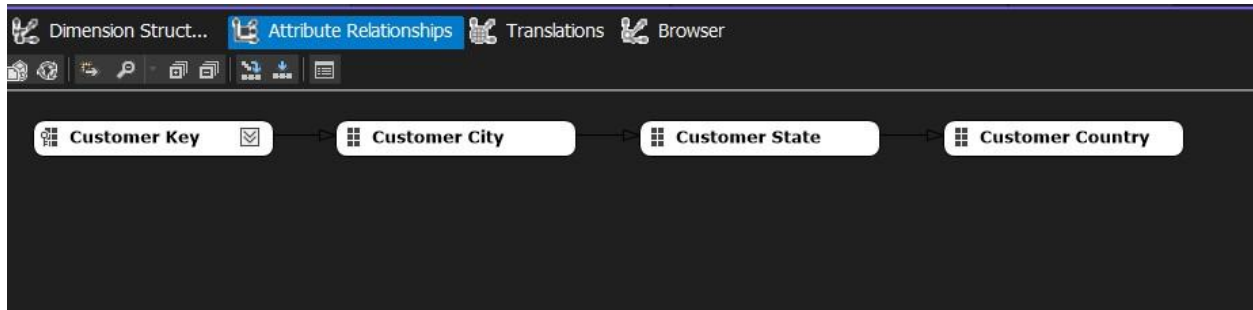
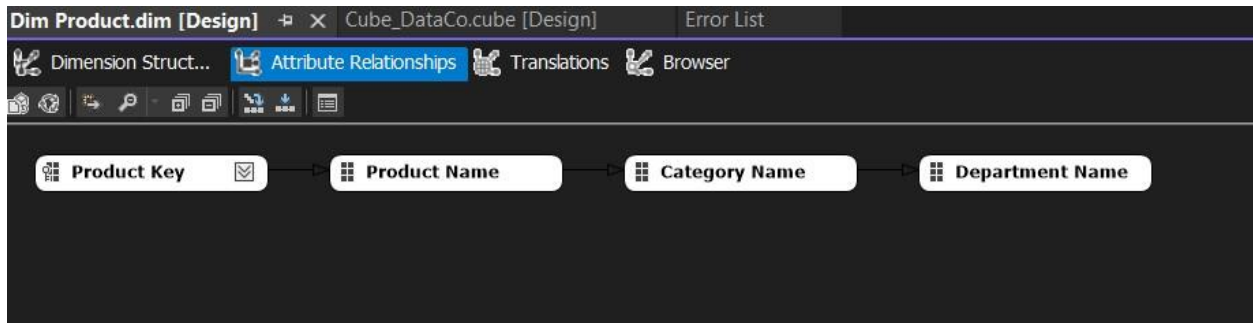


7. Created Hierarchies - Built user hierarchies:

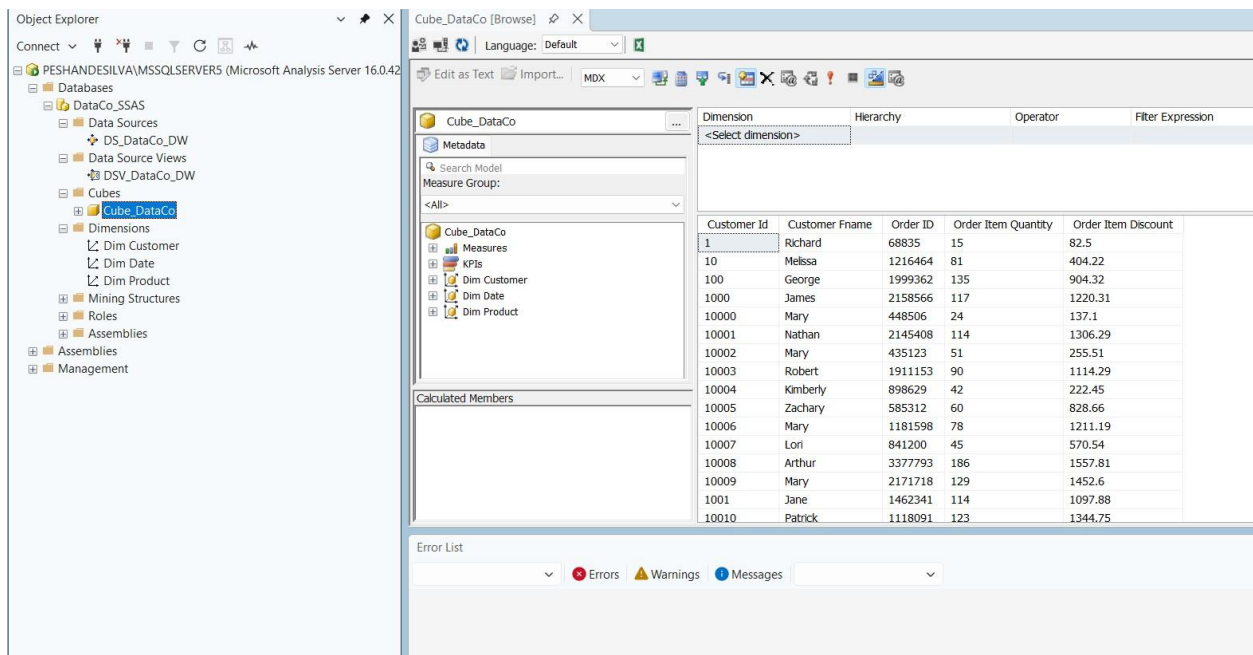
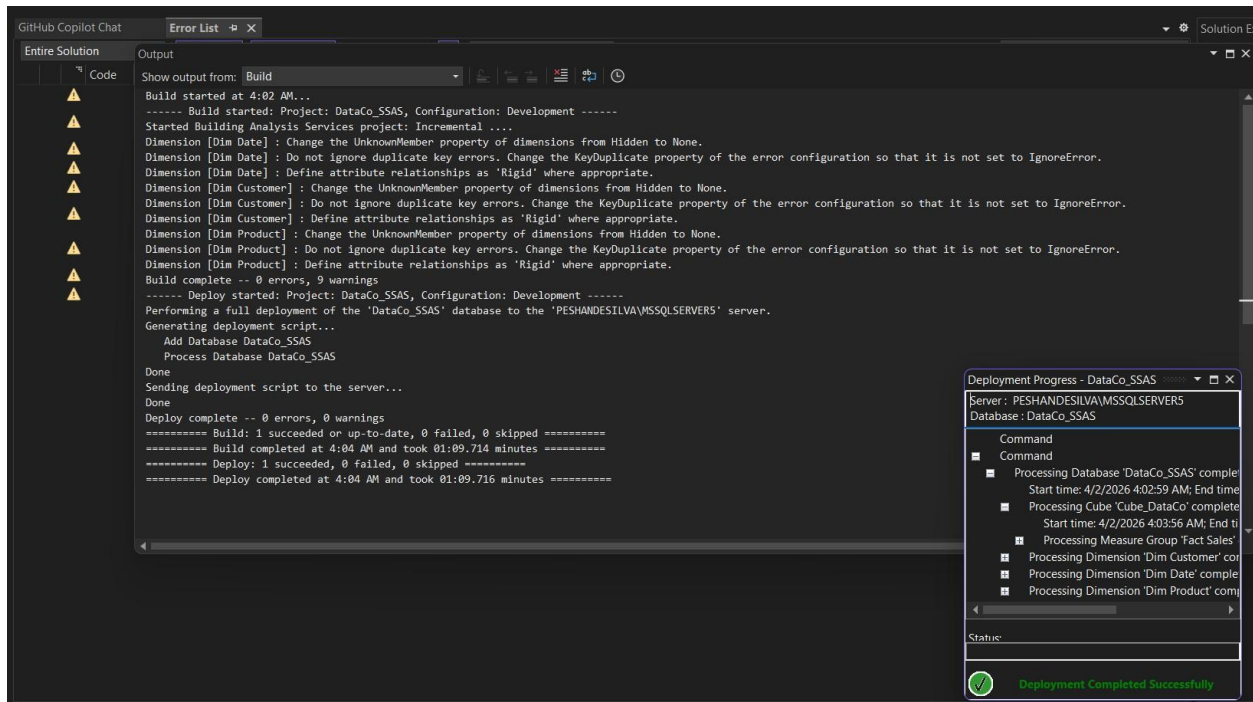
Department Hierarchy - Department Name – Category Name – Product Name

Customer Geography Hierarchy - Country – State – City

Date Hierarchy - Year→ Quater – Month – Date



8. Deployed the Cube - Configured deployment settings and successfully processed the cube to the SSAS server.
9. Validated Deployment - Confirmed cube accessibility through both SSDT browser and SQL Server Management Studio.



Step 3: Demonstration of OLAP operations

1. Connected Excel to SSAS Cube – Used Excel’s Get Data feature to establish a connection to the deployed SSAS cube.
2. Created PivotTable – Generated a PivotTable report in Excel linked to the cube for OLAP analysis.
3. Demonstrated Roll-up

Hierarchy Used: Date Hierarchy (Year level) and Product Hierarchy (Department level)

Measure Used: Order Item Quantity, Order Item Discount, Fact Sales Count

Configuration: Rows: Date Hierarchy (Year) | Values: Order Item Quantity, Order Item Discount, Fact Sales Count | Columns: None | Filters: None

Row Labels	Order Item Quantity	Fact Sales Count	Order Item Discount
# 2015	362127	170361	3557076.3
# 2016	221718	104910	2113929.87
# 2017	304761	142260	2933176.62
# 2018	263631	124026	2585871.39
Grand Total	1152237	541557	11190054.18

4. Demonstrated Drill-down

Hierarchy Used: Date Hierarchy: Year > Quarter > Month > Day

Measure Used: Order Item Quantity, Order Item Discount, Fact Sales Count

Configuration: Rows: Date Hierarchy (starting at Year) | Values: Order Item Quantity, Order Item Discount, Fact Sales Count | Columns: None

Row Labels	Order Item Quantity	Order Item Discount	Fact Sales Count
2015			
4			
10			
1	39054	370854.78	18150
13	37440	365074.95	17769
15	37779	358308.84	17445
27	37266	397039.95	17604
11			
2	38364	364207.53	18069
28	37512	392290.8	17847
12			
12	37746	363161.43	17688
24	36834	369912.36	17466
31	21915	215705.22	10353
7	38217	360520.44	17970
2016			
8	221718	2113929.87	104910
2017	304761	2933176.62	142260
2018	263631	2585871.39	124026
Grand Total	1152237	11190054.18	541557

5. Demonstrated Dice Operation

Hierarchies Used: Customer Geography (Country = Puerto Rico) AND Product Hierarchy (Department = Fan Shop)

Measure Used: Order Item Quantity, Days For Shipping Real

Configuration: Rows: Product Price | Values: Order Item Quantity, Days For Shipping Real | Columns: Date Hierarchy (Year) | Filters: Customer Geography = Puerto Rico AND Product Hierarchy = Fan Shop

Row Labels	Order Item Quantity	Days For Shipping Real	Total Order Item Quantity	Total Days For Shipping Real
2015				
11.53	201	144	222	72
149.99	18	15	21	18
199.99	5550	3417	4740	4113
29.98	348	156	318	308
299.98	4872	3000	4203	3444
349.98	6	24	27	15
399.98	6405	3762	5250	4626
49.97	21054	12693	18168	15321
99.00	99	9	111	48
Grand Total	38553	23196	33057	27975

6. Demonstrated Slice Operation

Hierarchy Used: Customer Geography (Country level)

Measure Used: Order Item Quantity

Configuration: Rows: Date Hierarchy (Year) | Values: Order Item Quantity | Columns: Product Hierarchy (Department) | Filters: Customer Geography = Puerto Rico

Customer Geography

Puerto Rico

Order Item Quantity

Column Labels

Row Labels

Apparel

Book Shop

Discs Shop

Fan Shop

Fitness

Footwear

Golf

Health and Beauty

Outdoors

Pet Shop

Technology

Grand Total

2015

35970

51

420

38553

2670

18096

36147

42

7323

213

717

14020

2016

21993

150

711

23196

1611

10857

21096

105

4251

222

105

8429

2017

30384

111

588

33057

1440

15618

30552

204

6414

9

273

11865

2018

25488

162

606

27975

1509

12552

26769

75

4887

108

582

10071

Grand Total

113835

474

2325

122781

7230

57123

114564

426

22875

552

1677

44386

PivotTable Fields

Choose fields to add to report:

Search

Fact Sales

Days For Shipping Real

Days Scheduled

Fact Sales Count

Late Delivery Risk

Order ID

Drag fields between areas below:

Filters

Customer Geography

Columns

Product Hierarchy

Rows

Date Hierarchy

Values

Order Item Quantity

7. Demonstrated Pivot

Hierarchies Used: Date Hierarchy (Year) swapped with Customer Geography (Country)

Measure Used: Order Item Quantity

Before State: Rows: Date Hierarchy (Year) | Columns: Customer Geography (Country: EE.UU., Puerto Rico) | Values: Order Item Quantity

After State: Rows: Customer Geography (Country) | Columns: Date Hierarchy (Year) | Values: Order Item Quantity

Order Item Quantity	EE. UU.	Puerto Rico	Grand Total
2015	221925	140202	362127
2016	137421	84297	221718
2017	186111	118650	304761
2018	162918	100713	263631
Grand Total	708375	443862	1152237

	A	B	C	D	E	F	G	H	I	J	K	L	M
1													
2													
3	Order Item Quantity	Column Labels											
4	Row Labels	2015	2016	2017	2018	Grand Total							
5	EE, UU.	221925	137421	186111	162918	708375							
6	Puerto Rico	140202	84297	118650	100713	443862							
7	Grand Total	362127	221718	304761	263631	1152237							
8													
9													
10													
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26													
27													

PivotTable Fields

Choose fields to add to report:

Search

Fact Sales

Days For Shipping Real

Days Scheduled

Fact Sales Count

Late Delivery Risk

Order ID

Order Item Discount

Drag fields between areas below:

Filters

Columns

Date Hierarchy

Rows

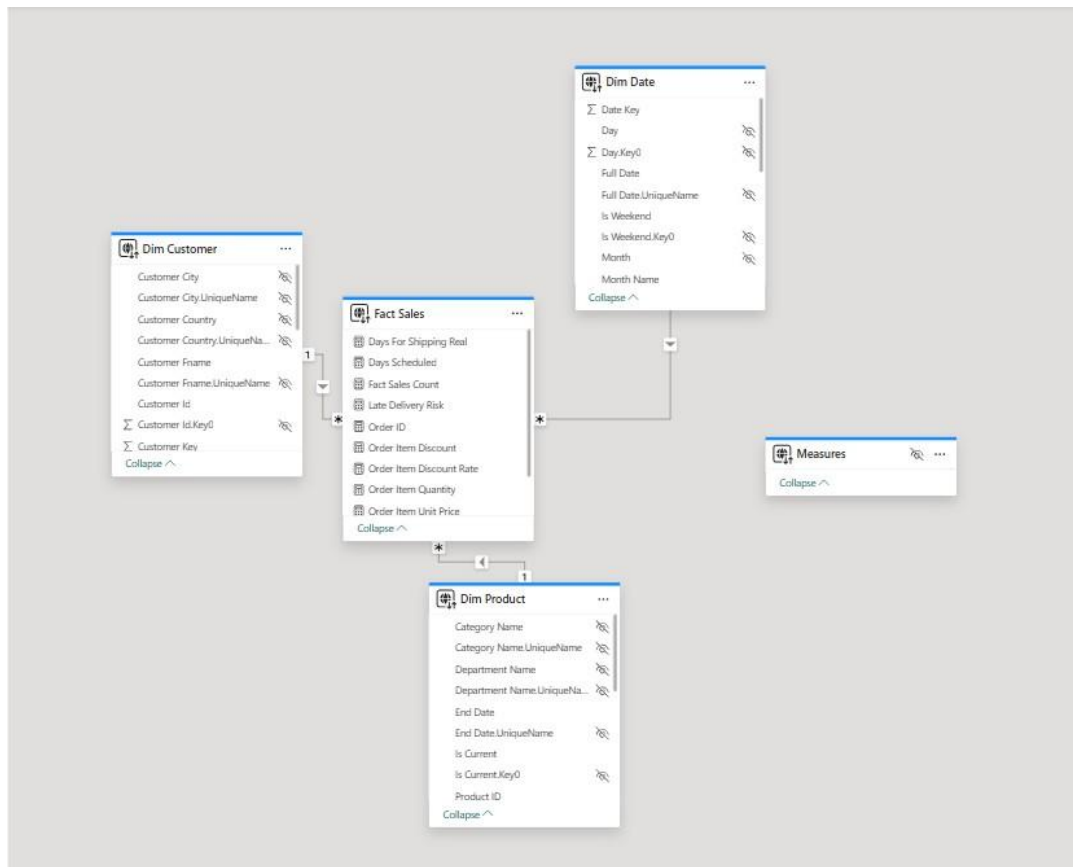
Customer Geography

Values

Order Item Quantity

Step 4: PowerBI Reports

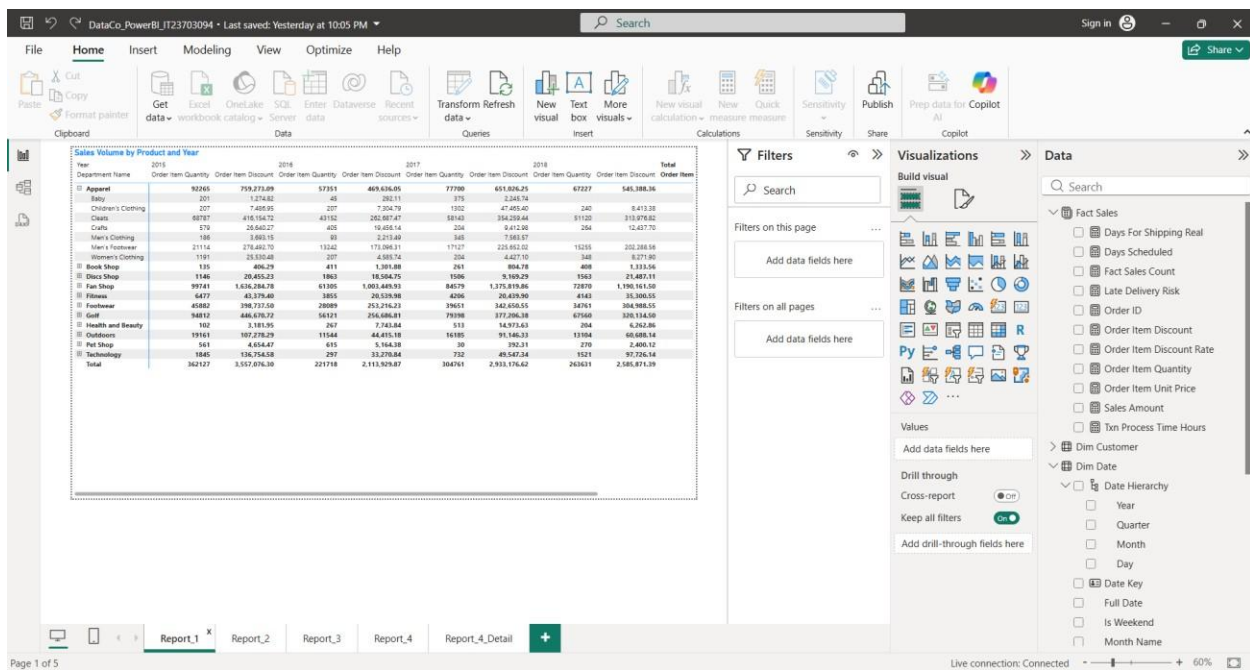
1. Connected to Data Source – Imported fact and dimension tables from the SQL Server Analysis Services Cube into Power BI Desktop.
2. The star schema was already linked with the fact table to the dimension tables using key columns since we uploaded the cube in the last step.



3. Report 1: Matrix Visual

a) Designed Matrix Layout:

- Added Hierarchical Rows → Department Name, Category Name
- And Columns → Year, Quarter
- And Values → Order Item Quantity, Order Item Discount



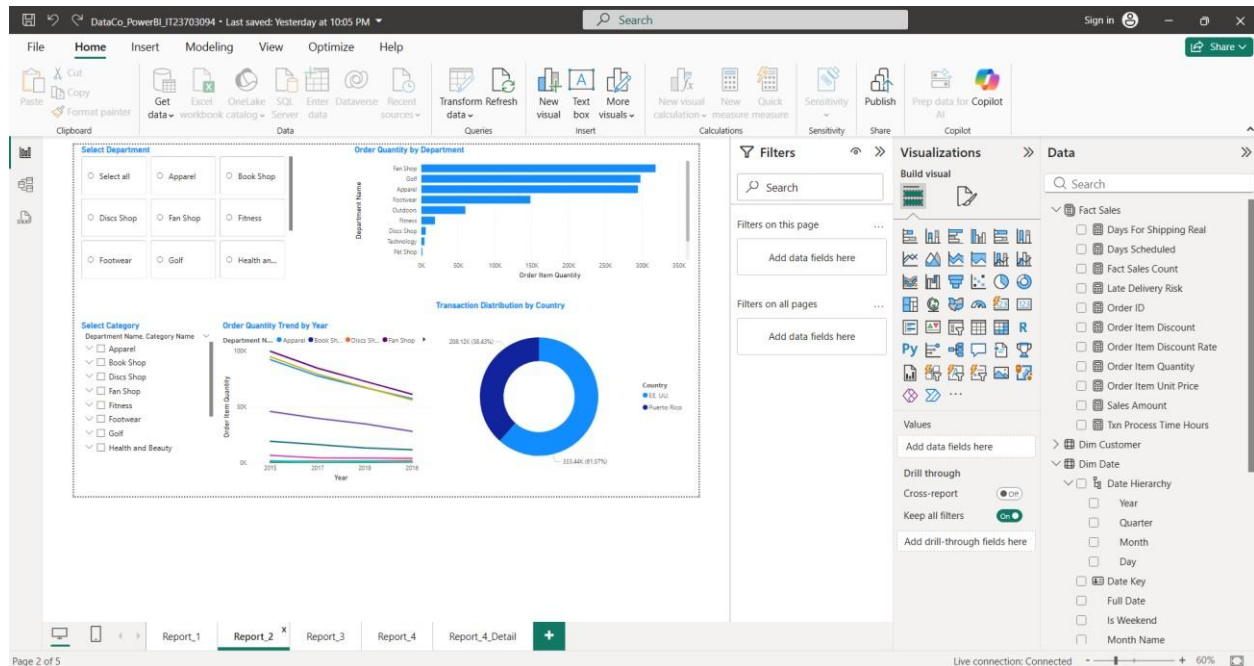
4. Report 2: Multiple Slicers with Cascading Filters

a) Created Interactive Cascading Filters

- Department Slicer: Added a list-style slicer for Department Name to serve as the primary filter.
- Category Slicer: Added a slicer for Category Name with Edit Interactions enabled, allowing the Department selection to dynamically filter the available Categories.

b) Added Supporting Visuals

- Bar Chart: Order Quantity by Department – Displays the sum of OrderItemQuantity across various departments.
- Line Chart: Order Quantity Trend by Year – Shows the performance of departments over time, using Year on the X-axis and OrderItemQuantity on the Y-axis.
- Donut Chart: Transactions by Country – Visualizes the distribution of total sales counts between EE, UU, and Puerto Rico.



5. Report 3: Drill-Down Report.

a) Built Hierarchical Visual

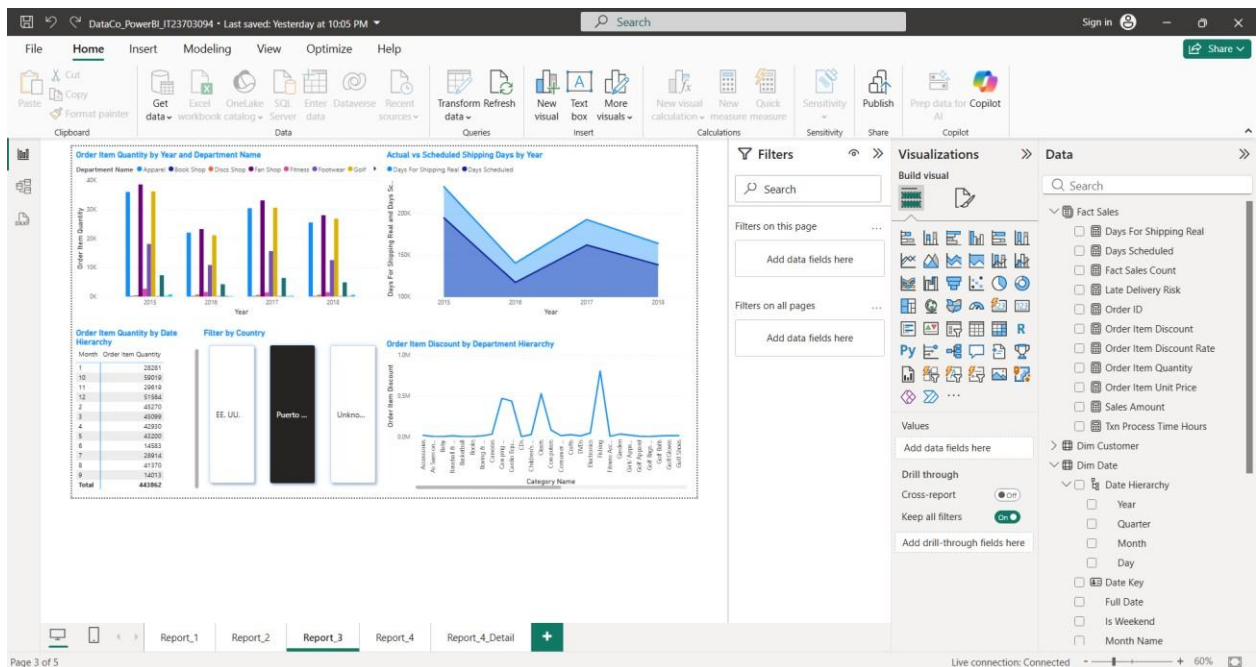
- Configured Clustered Column Chart: Created a multi-level visual by stacking the Date Hierarchy (Year, Quarter, Month, and Day) on the X-axis.
- Metric Integration: Set Order Item Quantity as the Y-axis value and categorized the data by Department Name in the Legend for granular comparison.

b) Configured Navigation Controls

- Drill-Down Setup: Enabled the standard Power BI drill-down arrows (Single Down, Double Down, and Up) in the visual header to allow users to navigate between different time grains.

c) Implemented Drill-Through Transitions

- Year to Quarter: Used the "Drill All Down" (double down arrow) functionality to transition the view from the Year level to the Quarter level.
- Quarter to Month: Executed a second drill-down to transition from Quarter to the Month level, as seen in the "Order Item Quantity by Date Hierarchy" matrix.
- Hierarchy Reset: Utilized the "Drill Up" arrow to return the visual to the high-level Year summary.



6. Report 4: Drill-Through Report

a) Built Summary Overview Page

- Primary Navigation Visual: Added a Clustered Bar Chart showing Order Quantity by Department, configured as the entry point for the drill-through action.
- Key Performance Indicator: Included a card visual for Total Transactions to provide a highlevel summary of the Fact Sales Count.

b) Created Target Detail Page

- Granular Reporting: Designed a separate "Report 4 - Detail" page featuring a comprehensive table with columns for Category, Product, Quantity, Discount, and Shipping Metrics.

- Trend & Card Visuals: Added a line chart to visualize the Order Trend by Category over time, alongside dedicated card visuals for filtered Order Quantity and Transaction Counts.

c) Configured Drill-Through Functionality

- Field Mapping: Set Department Name as the drill-through filter on the Detail page, which automatically generated the "Back" navigation button for seamless user flow.
- Dynamic Context: Configured the page to automatically filter all visuals (Table, Line Chart, and Cards) based on the specific Department selected from the Summary page.

d) Implemented User Navigation Test

- Action Verification: Verified the workflow by right-clicking a specific department bar (e.g., Fan Shop) on the Summary page and selecting Drill through → Report 4 - Detail to view the isolated data.

